

DAVID SALGADO JR.

Physics Graduate | Data & Technical Analysis | Python • SQL • Tableau

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SUMMARY

Physics graduate with an Astronomy concentration and Mathematics minor, hands-on experience analyzing experimental spectroscopy data, and a portfolio spanning Python, SQL, Tableau, APIs, web scraping, and scientific computing. Experienced in building reproducible workflows, validating quantitative results, visualizing findings, and communicating technical work through reports and presentations. Adaptable to data, research, technical operations, and automation-focused roles.

TECHNICAL SKILLS

Programming & Data: Python (pandas, NumPy, SciPy, matplotlib, Astropy), SQL, Jupyter Notebooks, Excel, Git/GitHub

Visualization & BI: Tableau Public, dashboards, data visualization, reporting

Automation & Data Collection: REST APIs, requests, BeautifulSoup, AWS fundamentals, scheduled workflows, logging

Analytical Methods: data cleaning, statistical analysis, uncertainty propagation, residual analysis, Gaussian fitting, calibration, basic machine learning (scikit-learn/KMeans)

Technical Communication: scientific writing, technical documentation, presentations, troubleshooting, training, teamwork, leadership

EXPERIENCE

Student Research Analyst - Physics Department | Appalachian State University | Boone, NC | Jun 2023 - Present

- Analyzed EUV spectroscopy data from highly charged ion research using Python and Jupyter; cleaned and calibrated datasets, fitted spectral lines, quantified uncertainty, and visualized trends across multiple beam energies.
- Applied Gaussian fitting, weighted averages, residual analysis, uncertainty propagation, and literature/theory comparisons to support identification and validation of spectral features.
- Contributed to analysis identifying 48 zirconium EUV lines, including 24 newly observed lines, and prepared figures, tables, and substantial text for a manuscript in preparation.
- Presented research at the 2024 APS DAMOP meeting and academic seminars; collaborated with research teams and participated in a research visit to the NIST EBIT facility.

SELECTED TECHNICAL PROJECTS

Data & Visualization Portfolio | GitHub | Remote | 2024 - Present

- Built public Python, SQL, and Tableau projects covering data cleaning, API retrieval, web scraping, clustering, scientific visualization, and dashboard development.

Private AWS/Python Automation Project | Independent | Remote | 2024 - Present

- Developing a private AWS/Python automation system using scheduled workflows, API integrations, cloud resources, logging, and operational debugging; proprietary implementation and business logic are intentionally kept private.

ADDITIONAL EXPERIENCE

Audio Visual Technician | Appalachian State University | Boone, NC | Dec 2022 - Jun 2024

- Troubleshot classroom and event audiovisual systems and provided clear technical guidance to faculty, staff, and students.
- Coordinated with team members to support reliable event execution in fast-moving environments.

Team Lead | Burger King | Dallas, NC | Aug 2020 - Aug 2022

- Trained new employees, supported shift productivity, and helped managers maintain operational flow.

EDUCATION & LEADERSHIP

Bachelor of Science in Physics, Astronomy Concentration; Mathematics Minor | Appalachian State University | Boone, NC | 2026

Associate of Science | Gaston College | Dallas, NC | 2021

Eagle Scout, Boy Scouts of America (2020) | National Youth Leadership Training (NYLT) (2018)